Fiscal Year:	FY 2008	**Task Last Updated:**	FY 07/16/2008
PI Name:	Hall, Eric J Ph.D., D.Sc.		
Project Title:	Individual Genetic Susceptibility		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	New York	**State:**	NY
Zip Code:	10032	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	05/19/2003	**End Date:**	06/01/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: Correct end date is 6/1/2009 per C. Guidry/JSC (2/2010) NOTE: Project to be extended into 2010, per J. Dardano; details of extension not yet available (7/2009) NOTE: Received NCE to 05/31/2009 per PI (7/2008) NOTE: Received NCE to 6/1/2008 per P. Denkins (6/07) NOTE: end date changed to 5/31/2007 per J. Dardano (5/06)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Brenner, David J (Columbia University) Smilenov, Lubomir (Columbia University)		
Grant/Contract No.:	NAG 9-1519		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The hypothesis upon which this proposal is based is that the human population is not homogeneous in radiosensitivity, but that radiosensitive sub-groups exist which would suffer an increased incidence of detrimental radiation effects, and distort the shape of the dose response relationship. The radiosensitivity of these groups depends on the expression levels of specific proteins. In a previous proposal funded by NASA we used knock-out mice to show unequivocally that haploinsufficiency for the ATM gene resulted in increased sensitivity to gamma-rays for both ocular cataracts (a deterministic effect) and oncogenic transformation in embryo fibroblasts (a stochastic effect). It would be extraordinarily difficult and expensive to conduct similar experiments with the high energy Fe-59 ions of principal interest to NASA, because of the need to breed a large number of animals, all of the appropriate age, to be available on the few occasions each year when Fe-59 ions are available at the Brookhaven Laboratory. It would be even more difficult and expensive to expand the study to include a number of other genes that are suspected of being involved with radiosensitivity, and to investigate the possible interaction of these various genes. We proposed instead to screen cells heterozygous for a number of genes; Atm, BRCA1, BRCA2 and rad9, as well as double heterozygotes for pairs of these genes. Wild type animals as well as heterozygotes and double heterozygotes will be exposed to graded doses of ?-rays at Columbia or Fe-56 heavy ions at Brookhaven National Laboratory, and thymocytes assayed for early apoptotic death. By this means we plan to discover which genes or combinations of genes, are important in conferring radiosensitivity. For genes, or combinations of genes that look promising for this apoptotic endpoint, more extensive studies would be carried out with the more relevant endpoint oncogenic transformation using the embryo fibroblasts. This strategy is based on preliminary studies with gamma-rays and cell lines with a mutated gene (namely Atm) which show that a reduction in radiation-induced apoptosis is mirrored by an increase in oncogenic transformation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The dose limits proposed for astronauts in space flight are different from those on the ground, though the basic philosophy of limiting the life-time risk of stochastic effects to 3% is preserved. There are several reasons for the differences, including the limited number of individuals exposed, the practicality of shielding and the presence of more urgent risks. Nevertheless the purpose of radiation protection remains to prevent deterministic effects of clinical significance and limit stochastic effects to acceptable levels. The levels proposed are based on the assumption that the human population is uniform in its radiosensitivity. Using two endpoints, namely cataracts as an important and relevant deterministic effect and oncogenic transformation in cultured embyro cells as a surrogate for carcinogenesis, the most relevant stochastic effect, we have shown that mice heterozygous for the atm gene are more radiosensitive than wild type animals. 1 to 3% of the US population are AT heterozygotes, and it would not be prudent to subject such individuals to substantial doses of radiation.
Task Progress:	Missions of longer duration in space, such as visits to the moon or Mars, must inevitably involve appreciable radiation exposure to astronauts. It is therefore important to know if there are sub-groups in the human population that are sensitive to radiation, particularly to radiation induced cancer. During the past year we have investigated whether mutations in two high penetrance genes, present at low frequency in the human population, confer sensitivity to radiation. We used mice heterozygous for Atm or BRCA1, and by mating singly heterozygous mice obtained animals with mutations in both genes. The biological endpoints scored were (1) Oncogenic transformation in fibroblasts obtained from embryos; this is a surrogate for carcinogenesis, and (2) Apoptosis in thymocytes, as a measure of genomic instability. Mice were irradiated with 1Gev/amu Fe ions at Brookhaven national laboratory, and compared with unirradiated controls. The conclusions are as follows: 1) Under normal conditions there is no difference in either transformation or apoptosis between wild type and heterozygous animals. 2) After exposure to Fe ions, we observed reduced apoptosis and elevated transformation in animals heterozygous for one gene, with a larger effect in double heterozygotes. 3) These data show that, in this model system, there are at least 2 high penetrance genes that are present in small percentages of the human population, that confer radiosensitivity. In previous years we had already shown that a 3rd gene, namely Mrad9 likewise resulted in radiosensitivity. Taken together, these results imply that the few percent of humans that carry mutations in one or more of these genes are likely to be sensitive to radiation.
Bibliography Type:	Description: (Last Updated: 10/26/2023)
Articles in Peer-reviewed Journals	Smilenov LB, Zhou G, Mitchell SA, Ludwig T, Hall EJ. "Heterozygosity for Atm and Brca1 Influence the Balance between Cell Transformation and Apoptosis." Cancer Letters, submitted for publication, July 2008. , Jul-2008